

NexusOne AI Architecture Diagrams

Logical layers, Phase 9.24 flow and component relationships

Project reference document. Product capabilities must be verified through code review, testing, security assessment and deployment evidence before production claims are made.

1. Logical architecture

CHANNELS AND USERS

Operations | Engineering | Governance | Leadership | Knowledge teams

PRESENTATION LAYER

React / TypeScript pages, dashboards, forms, portfolio tables and lifecycle actions

API AND APPLICATION LAYER

Spring Boot controllers, validation, DTO contracts, services and exception handling

DOMAIN AND GOVERNANCE LAYER

Entities, lifecycle states, scoring, review, approval, rejection, publication and audit timestamps

DATA AND KNOWLEDGE LAYER

JPA repositories, specifications, MySQL persistence, enterprise knowledge records and relationships

INTEGRATION LAYER

Connectors to telemetry, delivery, ITSM, cost, security, risk, documents and other authoritative sources

2. Phase 9.24 flow

Source knowledge -> validate and normalize -> create knowledge record -> review -> approve or reject -> publish -> discover and reuse

3. Core component relationship

Component	Responsibility
React page	Data entry, metrics, portfolio display and lifecycle actions.
TypeScript service	HTTP requests and typed responses.
REST controller	Endpoint mapping and request validation.
Service	Business rules, score calculation and lifecycle enforcement.
Entity and repository	Persistence and retrieval.
Specification	Filtered search by keyword, priority and status.
Exception handler	Consistent validation, state and not-found responses.

4. Architecture controls

- Use explicit module-scoped CSS to prevent cross-page theme overrides.
- Use LONGTEXT for large narrative fields in MySQL-backed entities.
- Keep API paths and frontend base URLs consistent.
- Enforce lifecycle transitions in the service layer.
- Treat calculated scores as indicators that require underlying evidence.
- Protect production integrations with authentication, authorization and audit logging.